

Work Placement: Student report

Master in Research and Innovation for Computer-based Science and Engineering

Entity/Department: Departamento de Ingeniería de la Información y de las Comunicaciones

Supervisor name: Ivan Erill Sagales

Student (name and ID): Enoch Young Min Shin, 1767585

Time period when the work placement took place:

February 2026 – July 2026

Brief description of work done and main achievements

The objective of this work placement was to create and evaluate a method of detecting specific patterns in DNA sequences. The work done was bioinformatics and software development, creating a pipeline to analyze the sample of sequences to detect these patterns. Main achievements were completing the full pipeline in Python to successfully detect the desired patterns, generating several HTML reports compiling the outputs and results, manual data annotation to create training data for the analysis, and literature review to understand the state of the art in the field.

Personal experience and suggestions for improvement

It was a fine experience.

(Signature and date)

Enoch Shin – 23 June 2026